MODULE 28 · DOMAIN 2 - PROCESS

# Quality and cost of quality

Quality requirements, managing versus controlling quality, the cost of quality, control charts and Pareto, audits, and quality built in.

DOMAIN II - Process  EXAM WEIGHT 41%  MODULE 28 of 49

admin@sthithapragna.com

Quality is planned and built in. **It cannot be inspected in.**


admin@sthithapragna.com

# Quality or grade?

QUALITY

## Meets requirements

- Degree to which it fulfils requirements
- Defects mean low quality
- Low quality is always a problem
- 'It does what was agreed, reliably'

GRADE

## A category of features

- Level of features or specification
- Basic versus premium
- Low grade can be exactly right
- 'A basic phone, or a flagship'

THE TIEBREAKER

A low-grade product can be high quality. A premium product full of defects is low quality.

# Gathering quality requirements

- From **stakeholders**: what does 'good' mean to the people who will use and run it?
- From **standards and regulation**: industry, safety, accessibility, data protection.
- From the **organisation**: quality policy and procedures, which are process assets.
- Non-functional measures: reliability, performance, usability, durability.
- Make each one measurable, so it can be checked.

> ON THE EXAM
>
> Quality requirements that cannot be measured cannot be controlled. Turn 'robust' into a number.

# The quality management plan

| SECTION | COVERS |
| --- | --- |
| Standards and objectives | Which standards apply; measurable quality targets |
| Roles and responsibilities | Who plans, performs and checks quality work |
| Activities and tools | Reviews, testing, audits, the tools and techniques to use |
| Metrics | What is measured, how, and the tolerance for each |
| Continuous improvement | How lessons feed back into the process |

ON THE EXAM

The plan says how quality will be achieved and checked. It is tailored to the project's size and risk.

# Manage quality or control quality?

MANAGE QUALITY

## Is the process right?

- Also called quality assurance
- Focuses on the process
- Audits, process analysis, prevention
- Builds confidence that quality will be met

CONTROL QUALITY

## Is the product right?

- Focuses on deliverables
- Inspection, testing, measurement
- Finds defects before release
- Feeds scope validation

THE TIEBREAKER

Checking whether the team follows its process: manage quality, via audit. Checking a deliverable: control quality.

# The cost of quality

| CATEGORY | KIND | EXAMPLES |
|---|---|---|
| Cost of conformance | Prevention | Training, design reviews, standards, good tools |
| Cost of conformance | Appraisal | Testing, inspection, audits |
| Cost of non-conformance | Internal failure | Rework and scrap found before release |
| Cost of non-conformance | External failure | Warranty claims, recalls, lost customers, liability |

ON THE EXAM

Money spent on prevention is almost always cheaper than money spent on failure, especially failure customers see.

# Why prevention wins

- A common rule of thumb: a defect costs about **1** to prevent, **10** to fix before release, **100** after release.
- Early investment in reviews, training and standards reduces every later cost.
- Cutting prevention to save money usually moves the cost to failure, where it grows.
- External failure also damages reputation, which no budget line captures.

> ON THE EXAM
>
> Asked where to invest in quality? Prevention first. Asked what to cut? Not prevention.

# Quality and sustainability

- The outline pairs them: **manage cost of quality and sustainability**.
- Rework and scrap are waste: materials, energy and effort thrown away.
- Durable, repairable, energy-efficient products are both higher quality and more sustainable.
- Include sustainability measures in quality requirements and acceptance criteria.

> ON THE EXAM
>
> Good quality is often good sustainability. Every defect prevented is waste avoided.

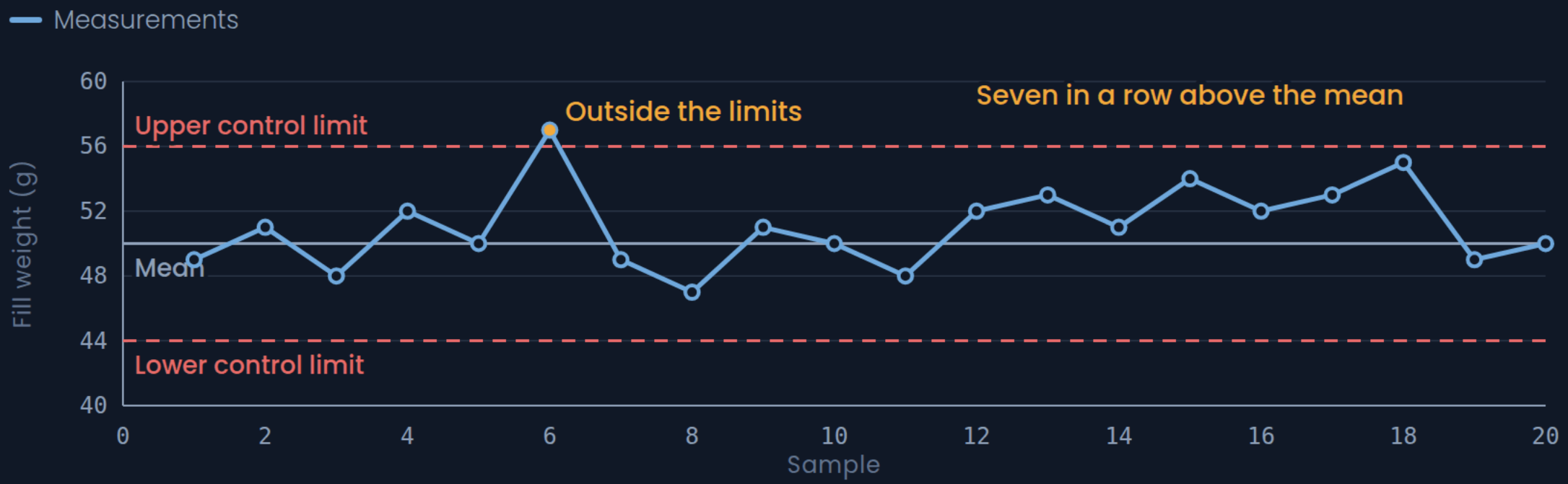
# Quality tools

| TOOL | USE IT TO |
| --- | --- |
| Checklist or check sheet | Make sure steps are done; tally defects by type |
| Cause-and-effect diagram | Explore possible causes of a problem, by category |
| Histogram | See how measurements are distributed |
| Pareto chart | Find the few causes behind most defects |
| Scatter diagram | See whether two variables are related |
| Control chart | Tell whether a process is stable over time |
| Flowchart | Map a process to find where defects arise |

ON THE EXAM

Match the tool to the question: which causes, which cause matters most, is the process stable?

# A control chart




ON THE EXAM

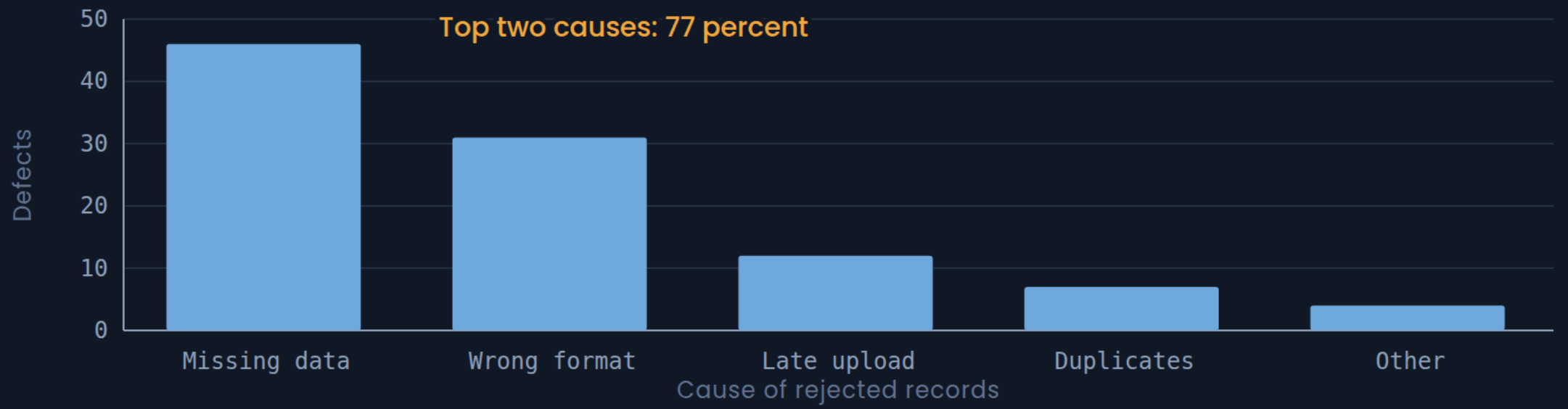
Out of control: a point outside the limits, or seven in a row on one side.

# Reading control charts

- **In control**: points vary randomly within the limits. Leave the process alone.
- **Out of control**: a point outside a limit, or seven consecutive points on one side of the mean.
- **Control limits** come from the process itself. **Specification limits** come from the customer.
- Adjusting a stable process after every normal variation, called tampering, makes it worse.

> ON THE EXAM
>
> Out of control means investigate for a special cause. In control means do not tamper.

# A Pareto chart of defect causes




ON THE EXAM

Sort causes from most to fewest. Fix the tallest bars first: the vital few behind most defects.

# Inspection, sampling and audits

- **Inspection** examines a deliverable against requirements. It is control quality.
- **Statistical sampling** inspects a representative sample when checking everything is impractical.
- A **quality audit** checks whether the project follows its quality processes and policies. It is manage quality.
- Audits also spread good practice and find process improvements.

> ON THE EXAM
>
> Inspection checks the product. An audit checks the process.

# Regulatory compliance in quality

- The outline asks you to **help ensure regulatory compliance** as part of quality.
- Identify the regulations that apply to the product: safety, medical, financial, accessibility.
- Build compliance checks into the definition of done and acceptance criteria.
- A deliverable that works but breaches regulation fails quality, however well it performs.

> ON THE EXAM
>
> Regulatory requirements are quality requirements. Module 33 covers compliance in full.

# Ongoing reviews and continuous improvement

- **Plan, do, check, act**: plan a change, try it, check the result, then standardise or adjust.
- Conduct **ongoing quality reviews**, not just one at the end.
- On agile teams, quality is built in: definition of done, test-first, continuous integration, pairing.
- Retrospectives are continuous improvement in a two-week rhythm.

> ON THE EXAM
>
> Quality improves in small, repeated cycles. Plan, do, check, act.

# Project Lantern: quality

- **Requirements**: bookings confirmed in under two seconds; accessible to screen readers; kiosks repairable on site.
- **Prevention**: design reviews with clinic staff, and test-first development on the app.
- **Control**: a control chart tracks booking response times; a Pareto chart ranks migration errors.
- **Audit**: a mid-project quality audit confirmed each clinic followed the go-live checklist.

> ON THE EXAM
>
> Measurable requirements, prevention first, charts for control, and an audit of the process.

# Scenario signals and the move

| THE SCENARIO SAYS | THE MOVE |
|---|---|
| Classify a quality cost | Prevention, appraisal, internal or external failure |
| Seven points in a row on one side of the mean | Investigate: the process is out of control |
| Is the team following its quality process? | Quality audit |
| Premium materials, but defects | Grade is not quality: it has a quality problem |
| Many defect types, limited time | Pareto: fix the vital few |
| Sponsor wants to cut reviews and training | Explain cost of quality: prevention is cheapest |

ON THE EXAM

**Build quality in, measure it, and spend on prevention before failure.**

A manufacturer reviews its quality costs for a new product line. They include staff training, design reviews, final testing, and warranty repairs for products that failed after customers bought them.

The finance director asks how the warranty repairs should be classified.

**Which cost of quality category do the warranty repairs belong to?**

A Prevention

B External failure

C Appraisal

D Internal failure

◆ PAUSE AND DECIDE

## Which cost of quality category do the warranty repairs belong to?

A **Prevention**

*Training and design reviews are prevention. Repairs are not.*

B **External failure**

Defects found after the customer has the product.

C **Appraisal**

*Final testing is appraisal. Repairs come after failure.*

D **Internal failure**

*Internal failure is caught before release. These reached customers.*

THE SIGNAL

Found before release: internal failure. Found by the customer: external failure, the most expensive.

> A control chart for a packaging line shows every point within the control limits. However, the last seven consecutive measurements all sit above the mean.
>
> The line supervisor says there is nothing to worry about because nothing is outside the limits.

## What should the project manager do?

A. Take no action, since all points are within the control limits.

B. Adjust the machine settings after each measurement to bring values back to the mean.

C. Widen the control limits so the process has more room.

D. Investigate for a special cause, since the run of seven indicates the process is out of control.

◆ PAUSE AND DECIDE

## What should the project manager do?

A No action.

*Misses the second signal. Seven in a row is not random variation.*

B Adjust after each measurement.

*Tampering. It adds variation rather than removing the cause.*

C Widen the limits.

*Control limits come from the data, not preference. It hides the signal.*

D Investigate for a special cause.

The rule of seven: a run on one side means something has shifted.

THE SIGNAL

Within the limits is not the whole test. Seven in a row on one side also means out of control.

> Final inspections keep finding the same kinds of defects in a construction project's electrical work. The project manager suspects the crews are not following the agreed installation procedures.
>
> The project manager wants to find out whether that is true.

**Which activity should the project manager use?**

A. A quality audit of the installation process.

B. More final inspections of completed work.

C. A Pareto chart of the defect types.

D. A control chart of defects per week.

◆ PAUSE AND DECIDE

## Which activity should the project manager use?

**A** Quality audit.

Checks whether the process and procedures are being followed. Exactly the question.

**B** More inspections.

*Finds more defects in the product; says nothing about the process.*

**C** Pareto chart.

*Ranks defect types, but cannot show whether procedures are followed.*

**D** Control chart.

*Shows stability over time, not compliance with procedures.*

THE SIGNAL

'Are they following the process?' is an audit question. Inspection checks the product.

A client ordered a basic, low-cost staff canteen fit-out. The contractor used premium fittings the client had not asked for, and the handover inspection found several leaking taps and faulty lights.

The contractor argues that the fit-out is high quality because the fittings are premium.

## Which statement is correct?

A The fit-out is high quality, because premium fittings were used.

B The problem is the low grade the client chose.

C The fit-out has a quality problem, the defects, and an unrequested change of grade.

D The fit-out is acceptable, because premium fittings outweigh minor defects.

◆ PAUSE AND DECIDE

## Which statement is correct?

A High quality because premium.

*Confuses grade with quality. Defects mean low quality.*

B Low grade was the problem.

*Low grade was agreed and is fine. It did not cause the leaks.*

C A quality problem, plus an unrequested grade.

Defects are low quality; premium fittings nobody asked for are gold plating.

D Acceptable overall.

*Defects fail the requirements, whatever the grade.*

THE SIGNAL

Premium is grade, not quality. Defects are always a quality problem.

> A data migration has produced four hundred rejected records across eleven different error types. The team has one week to reduce rejections before the next migration run.
>
> They cannot fix every error type in that time.

## Which approach should the project manager use first?

A. A Pareto analysis, to find the few error types causing most rejections.

B. Fix all eleven error types equally, a little at a time.

C. A scatter diagram of rejections against record size.

D. A cause-and-effect diagram for each of the eleven error types.

◆ PAUSE AND DECIDE

## Which approach should the project manager use first?

**A** Pareto analysis.

Finds the vital few error types, so a limited week goes where it matters most.

**B** Fix all equally.

*Spreads effort thinly; the biggest causes get the same attention as the smallest.*

**C** Scatter diagram.

*Tests one possible relationship; does not prioritise causes.*

**D** Eleven fishbone diagrams.

*Useful later for the top causes. Too slow as a first step.*

THE SIGNAL

Many causes, limited time: Pareto first. Then dig into the biggest causes.

> To save money, the sponsor proposes cancelling the planned design reviews and staff training, and relying on thorough final testing instead.
>
> The product is a medical device, where defects found by patients would be very costly.

## What should the project manager do?

A Agree, since final testing will catch any defects.

B Agree, and reduce final testing as well to save more.

C Explain the cost of quality, showing that prevention is cheaper than failure, and recommend keeping the reviews and training.

D Agree, and hire extra inspectors to compensate.

◆ PAUSE AND DECIDE

## What should the project manager do?

A Rely on final testing.
*Testing finds defects late, when fixing them costs far more.*

B Cut testing too.
*Removes both prevention and appraisal on a medical device. Dangerous.*

C Explain cost of quality; keep prevention.
Prevention is cheapest, and external failure here could harm patients.

D More inspectors.
*Swaps cheap prevention for expensive appraisal. Quality cannot be inspected in.*

THE SIGNAL

Cutting prevention moves cost into failure, where it grows. Advise the sponsor with cost of quality.

Build quality in. **Prevent** before you inspect.
Read the chart before you react.


admin@sthithapragna.com